Linear Algebra **Unit 5: Determinants and Linear Transformations**

Practice Test — Study Copy

Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Determinant _______
2. Minor _______
3. Cofactor expansion _______
4. Linear transformation _______
5. Orientation _______
6. Singular matrix _______
7. Area/volume factor _______
8. Product rule _______

(A) The smaller determinant left after crossing out one entry's row and column.

(B) A map $\mathbf{x} \mapsto A\mathbf{x}$ that fixes the origin and sends grid lines to straight, evenly spaced lines.

(C) A matrix with $\det = 0$: its columns are dependent, it collapses space, and it has no inverse.

(D) A number computed from a square matrix; its absolute value is the area (or volume) of the box built from the columns.

(E) The rule $\det(AB) = \det A \det B$: composing transformations multiplies their scaling factors.

(F) Computing a determinant along one row: each entry times its signed minor, with the pattern $+ - + \dots$

(G) Whether a transformation keeps or flips the handedness of space — kept when $\det > 0$, flipped when $\det < 0$.

(H) The quantity $|\det A|$, the number every area (or volume) is multiplied by under the transformation A.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. The determinant of the identity matrix I is (A) 0 (B) 1 (C) n (D) undefined
2. Swapping two rows of a matrix (A) leaves det unchanged (B) negates det (C) doubles det (D) makes $\det = 0$
3. Adding a multiple of one row to another row (A) negates det (B) scales det (C) leaves det unchanged (D) makes $\det = 0$
4. A square matrix with $\det = 0$ is (A) invertible (B) singular (C) orthogonal (D) symmetric
5. The number $|\det A|$ tells you (A) the trace (B) how much A scales area/volume (C) the rank (D) nothing geometric
6. The determinant of a product satisfies (A) $\det(AB) = \det A + \det B$ (B) $\det(AB) = \det A \det B$ (C) $\det(AB) = \det A - \det B$ (D) no rule

Part C — Short Answer & Computation (35 pts)

Show your work.

1. Compute the determinant of $A = \left[\begin{smallmatrix} 1 & 3 \\ 2 & 1 \end{smallmatrix}\right]$. Then give the **area** of the parallelogram built from its columns, and say whether A **preserves or reverses orientation**.

2. Compute the determinant of $B = \left[\begin{smallmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{smallmatrix}\right]$ by **cofactor expansion** across the top row (show the three 2×2 minors).

3. Compute the determinant of $C = \left[\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 0 & 2 & 5 \end{smallmatrix}\right]$ by **reducing to triangular form** (elimination), then multiplying the pivots.

4. A 3×3 matrix A has $\det A = 6$. Find each new determinant and name the row rule you used: **(a)** two rows of A are swapped; **(b)** one row of A is multiplied by 3; **(c)** $2 \times$ (row 1) is added to row 3; **(d)** $\det(2A)$.

5. Two 3×3 matrices have $\det A = 4$ and $\det B = 2$. Use the product rule to find **(a)** $\det(AB)$, **(b)** $\det(A^{-1})$, **(c)** $\det(A^{\mathsf{T}})$, **(d)** $\det(A^2)$.

6. Photo software applies the transformation $\mathbf{x} \mapsto A\mathbf{x}$ with $A = \left[\begin{smallmatrix} 3 & 1 \\ 0 & 2 \end{smallmatrix}\right]$. **(a)** What is the area of the image of the unit square? **(b)** A triangle has area 4; what is the area of its image? **(c)** Does the image keep its orientation?

7. Let $M = \left[\begin{smallmatrix} 2 & 1 \\ 4 & 2 \end{smallmatrix}\right]$. **(a)** Compute $\det M$. **(b)** Is M invertible? **(c)** Describe what the transformation $\mathbf{x} \mapsto M\mathbf{x}$ does to the unit square (what does it collapse to, and why).

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. You compute the determinant of a 3×3 matrix, then perform one elimination step — adding a multiple of one row to another — and compute the determinant again. **(a)** Explain why the determinant does *not* change. **(b)** Using the picture of the box (parallelepiped) built from the rows, explain why sliding the box sideways this way keeps its volume the same.

2. A transformation $\mathbf{x} \mapsto A\mathbf{x}$ sends the unit square to a parallelogram whose sides are the columns $A\mathbf{e}_1$ and $A\mathbf{e}_2$. **(a)** Explain why the area of that parallelogram is $|\det A|$, and why this makes $|\det A|$ the factor by which *every* area is scaled. **(b)** Explain why $\det A = 0$ forces A to have no inverse, using what happens to the unit square.